

Chip is the leading provider of femto cell chips, and once more demonstrates this position of leadership with its software-defined architecture," says Stuart Carlaw, ABI Research.

Femto cells are tiny, low-power 3G radio systems that plug into a residential broadband connection to provide a mobile signal directly in the home. A femto cell is smaller than a picocell and used to describe a very small radio cell associated with a cellular radio base station located in a home or small office.

Femto cells: Ideal for indoor coverage

Cellular radio is the fastest growing segment of the communications industry. Cellular companies reported a subscription base of more than 200 million people in 1997. This figure grows by an average of 150,000 new subscribers every day. With about 30 per cent of mobile calls made at home, operators with coverage holes in residential areas have the potential to boost their revenues significantly by deploying femto cells.

A survey in June 2008 predicted that 50 per cent of phone calls and 70 per cent of data calls would take place indoors in coming years. Therefore indoor coverage with high QoS (quality of services) will be in demand. Since other cells' coverage for such huge demands of indoor access will be expensive, the judicious choice will be to provide the service with femto cells.

Low-powered base stations in femto cells will provide desired solutions for indoor coverage. These will change the way operators build their networks and grow capacity and coverage.

The use of femto cells will benefit both users and operators in several aspects:

1. Communication with large capabilities and throughputs
2. Better signal quality due to proximity between the transmitter and the receiver in the smallest cell
3. Power savings
4. Reduced electromagnetic interference
5. Faster data speed
6. Low-cost service
7. Improved indoor coverage of both voice and data services

The transmit power of the femto base station, to be placed in individual homes, will be low. The femto cell connects to the service provider's network over the household's existing broadband link such as conventional digital subscriber line (DSL) or cable broadband access; current designs typically support two to four active mobile phones in a residential setting.

Femto cells allow service providers to extend service coverage indoors, especially where access would otherwise be limited or unavailable. Cost saving is a primary drive for femto services for users. The cost savings can be passed on to customers; for example, via femtozone tariffs, making

broadband-bundled mobile price plans competitive not only with the fixed-line telephone but also with the TV and PC for entertainment and information services in the home.

Femto cells help in two ways:

- Removing indoor data sessions from the macro network reduces the number of users each macro cell needs to support.
- Because of the way CDMA works, if indoor users are served via femto cells instead of the macro cell, the capacity of the macro network increases in proportion to the number of users who have been removed from the cell.

Fixed-mobile convergence possible

Over the years, we have seen users ignoring landline in favour of mobiles. Clearly, there is a need to hybrid the fixed line with mobile.

Femto cells are an alternative way to deliver the benefits of fixed and mobile convergence. A femto cell-based deployment will work with existing handsets but requires installation of a new access point that uses licenced spectrum.

Femto cells provide a solution to a hybrid cellular and broadband network by using an existing broadband link to connect securely to wireless and then send and receive cellular signals throughout a home or office. ●

The author is a regular contributor to EPT

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